

The empirical recurrence for OEIS A263435: recovering a conjecture that is not written down

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Abstract

OEIS A263435 records “Empirical recurrence of order 88”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 169$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 88, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 88 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A263435 is “Number of $(n+1) \times (2+1)$ 0..4 arrays with each row and column divisible by 13, read as a base-5 number with top and left being the most significant digits.”. It has offset 1 and begins

2, 21, 180, 1225, 9820, 90785, 841164, 7551453, ...

The entry states:

Empirical recurrence of order 88 (see link above)

As of the “Last modified” line on the live entry (Oct 18 2015 10:54:01, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 169$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 169$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 338$ exact terms of the model returns order 88 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{88} c_i a(n-i),$$

c_1	12	c_{23}	−2854810695620064	c_{45}	−998868527396960375706720	c_{67}	1484
c_2	−24	c_{24}	−149190499449337	c_{46}	1997737054793920751413440	c_{68}	−2425
c_3	48	c_{25}	58886499905793324	c_{47}	−3995474109587841502826880	c_{69}	−594
c_4	4533	c_{26}	−117772999811586648	c_{48}	5801155588129077747028003	c_{70}	1188
c_5	−55356	c_{27}	235545999623173296	c_{49}	10295615134207897092201564	c_{71}	−2376
c_6	110712	c_{28}	−118169477118354447	c_{50}	−20591230268415794184403128	c_{72}	3915
c_7	−221424	c_{29}	−3292886267043212556	c_{51}	41182460536831588368806256	c_{73}	5492
c_8	−5018230	c_{30}	6585772534086425112	c_{52}	−62275853274350597253424348	c_{74}	−1098
c_9	64647240	c_{31}	−13171545068172850224	c_{53}	−76338971444424600335032944	c_{75}	2196
c_{10}	−129294480	c_{32}	11268243475925659405	c_{54}	152677942888849200670065888	c_{76}	−3642
c_{11}	258588960	c_{33}	128211979652349091620	c_{55}	−305355885777698401340131776	c_{77}	−226
c_{12}	1780399901	c_{34}	−256423959304698183240	c_{56}	475342961907066862403959254	c_{78}	4536
c_{13}	−26536578012	c_{35}	512847918609396366480	c_{57}	403002172669165677955124472	c_{79}	−907
c_{14}	53073156024	c_{36}	−559903112156581543940	c_{58}	−806004345338331355910248944	c_{80}	15114
c_{15}	−106146312048	c_{37}	−3538121026308948802320	c_{59}	1612008690676662711820497888	c_{81}	688
c_{16}	−290865419748	c_{38}	7076242052617897604640	c_{60}	−2562799156383289395144500043	c_{82}	−137
c_{17}	5613311277936	c_{39}	−14152484105235795209280	c_{61}	−1486583936933781494675957244	c_{83}	275
c_{18}	−11226622555872	c_{40}	17754583851556341929321	c_{62}	2973167873867562989351914488	c_{84}	−459
c_{19}	22453245111744	c_{41}	69994675886039801033748	c_{63}	−5946335747735125978703828976	c_{85}	−5
c_{20}	22053147639178	c_{42}	−139989351772079602067496	c_{64}	9601204775509101033767398201	c_{86}	10
c_{21}	−713702673905016	c_{43}	279978703544159204134992	c_{65}	3712257648593307168867801108	c_{87}	−2
c_{22}	1427405347810032	c_{44}	−383392128623851975582760	c_{66}	−7424515297186614337735602216	c_{88}	35

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 88, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $88 + 4$ terms removed returns order 88 as well, so $\deg q_{\min} = 88 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $88 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 88 that this sequence satisfies — and that family turns out to have exactly one member.

References

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